Scalability & Future Plan

Date	11 July 2026
Team ID	XXXXXX
Project Name	Rising waters- Flood Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Prediction accuracy depends on dataset quality	Wrong or less accurate predictions may occur	High
2	Supports only limited flood-related input parameters	Prediction capability is limited	medium
3	Project runs mainly in Google Colab	Not easily accessible as a web application	medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports single/local execution	Deploy as a web application to support multiple users
2	Data Storage	Dataset stored locally	Integrate cloud database for storing large datasets
3	Performance	Suitable for small datasets	Optimize the model and use larger datasets for faster and better predictions
4	Security	Basic project security	Add user authentication and secure cloud deployment

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2	Improve prediction accuracy using larger and updated datasets	1-2 months	More accurate flood predictions
Phase 3	Develop a web application for public access	2-3 months	Easy access for multiple users
Phase 4	Integrate real-time weather and rainfall	3-6 months	Real-time flood forecasting